

CSCB07 – Software Design

Tutorial Exercise #1

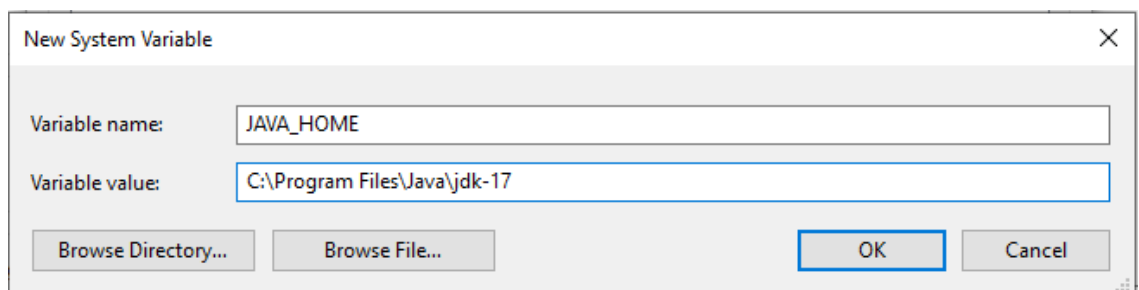
Objectives

- Getting familiar with Java
- Learning how to compile and run a Java program from a command line
- Getting familiar with Git

Instructions

1. Download and install JDK:
<https://www.oracle.com/java/technologies/javase/jdk17-archive-downloads.html>
2. Windows users might have to set the JAVA_HOME environment variable as follows:

Control Panel -> System and Security -> System -> Advanced system settings -> Advanced -> Environment Variables -> System Variables -> New and set JAVA_HOME by specifying the path to the JDK directory (the path on your machine might be different than the one shown in the example below)



3. If Git is not installed on your machine, [download](#) it and then install.
4. If you don't have a Github account, create one [here](#).
5. Create a personal access token to be used for authentication. More information could be found [here](#).
6. On Github, create a repo named **b07lab1** as per these [instructions](#) and make sure to make it public.
7. Navigate to the repo and create a new file: name it "info", add some information about the lab, and click the "Commit new file" button.
8. On your machine, do the following steps using Command Prompt/Terminal:
 - a. Clone the repo using the command **git clone RepoURL** (*RepoURL* should be replaced by the actual URL of your repo)
 - b. **cd b07lab1**
 - c. Windows users:
notepad Polynomial.java (answer "Yes" when asked if you want to create a new file)

Mac users:

touch Polynomial.java

open Polynomial.java

- d. Develop class *Polynomial* as follows:
- It has one field representing the coefficients of the polynomial using an array of *double*. A polynomial is assumed to have the form $a_0 + a_1x^1 + \dots + a_{n-1}x^{n-1}$. For example, the polynomial $6 - 2x + 5x^3$ would be represented using the array [6, -2, 0, 5]
 - It has a no-argument constructor that sets the polynomial to zero (i.e. the corresponding array would be [0])
 - It has a constructor that takes an array of *double* as an argument and sets the coefficients accordingly
 - It has a method named **add** that takes one argument of type *Polynomial* and returns the polynomial resulting from adding the calling object and the argument
 - It has a method named **evaluate** that takes one argument of type *double* representing a value of x and evaluates the polynomial accordingly. For example, if the polynomial is $6 - 2x + 5x^3$ and **evaluate(-1)** is invoked, the result should be 3.
 - It has a method named **hasRoot** that takes one argument of type *double* and determines whether this value is a root of the polynomial or not. Note that a root is a value of x for which the polynomial evaluates to zero.
- e. Save the file
- f. Windows users:
- notepad Driver.java** (answer "Yes" when asked if you want to create a new file)

Mac users:

touch Driver.java

open Driver.java

- g. Add the following code, then save and close the file

```
public class Driver {
    public static void main(String [] args) {
        Polynomial p = new Polynomial();
        System.out.println(p.evaluate(3));
        double [] c1 = {6,0,0,5};
        Polynomial p1 = new Polynomial(c1);
        double [] c2 = {0,-2,0,0,-9};
        Polynomial p2 = new Polynomial(c2);
        Polynomial s = p1.add(p2);
        System.out.println("s(0.1) = " + s.evaluate(0.1));
        if(s.hasRoot(1))
            System.out.println("1 is a root of s");
        else
            System.out.println("1 is not a root of s");
    }
}
```

- h. **javac Driver.java**
- i. **java Driver**
- j. Verify that the output you obtain is as follows:
0.0
s(0.1) = 5.8041
1 is a root of s
- k. **git add *.java**
- l. **git commit -m "added Polynomial.java and Driver.java"**
- m. **git push**

9. Submit the URL of your repo to "Exercise 1" on Quercus.